

PROJECT MANAGEMENT

UNIT-3

[CS-604]

SYLLABUS-

Software management Disciplines, Iterative process planning. Project organisations and responsibilities. Process automation. Project control And process instrumentation - core metrics, management indicators, life cycle expectations. Process discriminants.

[May 2023]

[May 2024]

[June 2025]

ITERATIVE PROCESS PLANNING

Iterative Process Planning is a project management approach in which the project is divided into small repeated phases or cycles called iterations.

Each iterations performs planning, development, testing & evaluation of a small part of the project.

The output of one iteration becomes the input for the next iteration.

Each iteration includes:

Planning

Designing

Developing

Testing

Reviewing

Improving

After completing one cycle, the team learns from the results & plans the next cycle better.

Example-

Suppose you are making a shopping website. Instead of building the full website at once:

Iteration-1:

- Create homepage
- Add login page
- Basic testing

Iteration-2:

- Add cart system
- Improve design
- Fix bugs

Iteration - 3:

- Add payment gateway
- Add search feature
- Optimize speed

So, the product improves step-by-step.

Need of Iterative Process-

Traditional planning has many problems:

- Difficult to change requirements
- Bugs found very late
- Customers can't see progress early.
- High risk of project failure.

Iterative planning solves these problems because:

- Work is done in small steps
- Testing happens continuously
- Changes can be added anytime
- Customers stay involved.

Main Characteristics of Iterative Process -

1. Repeated Development Cycles -

The project is divided into many small cycles.

Each cycle contains:

- Planning
- Designing
- Coding
- Testing
- Reviewing

This cycle repeats until the project finishes.

2. Incremental Improvement -

Each iteration improves the previous version.

Example:-

Version 1 → Basic features

Version 2 → More features

Version 3 → Better Performance

3. Continuous Testing -

Testing is done after every iteration.

Benefits:

- Bugs are found early
- Easier to fix errors
- Better product quality

4. Customer Feedback -

Customers can review each version and suggest improvements.

This helps:

- Understanding user needs
- Improving satisfaction
- Reducing misunderstandings

5. Flexible Planning -

Requirements can change during development.

This flexibility is very useful in:

- Software projects
- Mobile apps
- Websites
- AI projects

6. Risk Reduction-

Small iterations reduce project risk because:

- Errors are detected early.
- Problems are fixed quickly
- Failure chances decrease.

Advantages

1. Easy to Handle Changes-

Requirements can change during development.

Example:

If customers suddenly want:

- Dark mode
- AI chatbot

These can be added in future iterations

2. Early Detection of Errors-

Since testing happens regularly:

- Bugs are found early
- Fixing becomes easier

3. Better Product Quality

Continuous improvement increases quality.

Each version becomes better than the previous one.

4. Customer Satisfaction-

Customers can-

- See progress regularly
- Suggest changes
- Get early versions.

5. Faster Delivery-

A basic working product can be delivered early.

Customers do not need to wait for the full projects.

Applications-

- Artificial Intelligence Projects
- Research Projects
- Product Development

PROJECT ORGANIZATIONS & RESPONSIBILITIES ^[June 2025]

A project cannot succeed only because of technology or money.

A project becomes successful when:

- People work together properly.
- Responsibilities are clearly defined.
- Communication is effective.

To manage all these things, a proper Project Organisation structure is needed.

Project organizations defines:

- who will do the work.
- who will manage the work.
- who will make decisions
- who is responsible for project process.

Project Organization is the structure that defines:

- Roles
- Responsibilities
- Authority
- Communication relationships

Basically among the people involved in a project.

It helps in coordinating project activities effectively.

Basically, Arranging people & assigning responsibilities in a systematic way to complete the project successfully.

Importance of Project Organization -

Project organisation is important because it:

- Clarifies responsibilities
- Improves communication
- Reduces confusion
- Increases efficiency
- Helps in proper decision-making
- Improves coordination
- Ensures accountability

Without proper organizations -

- Work may overlap
- Tasks may be missed
- Team conflicts may occur
- Deadlines may fail.

Types of Project Organization Structures -

There are mainly three types:

1. Functional organization
2. Projectized Organization
3. Matrix Organization

1. Functional Organization -

In this structure:

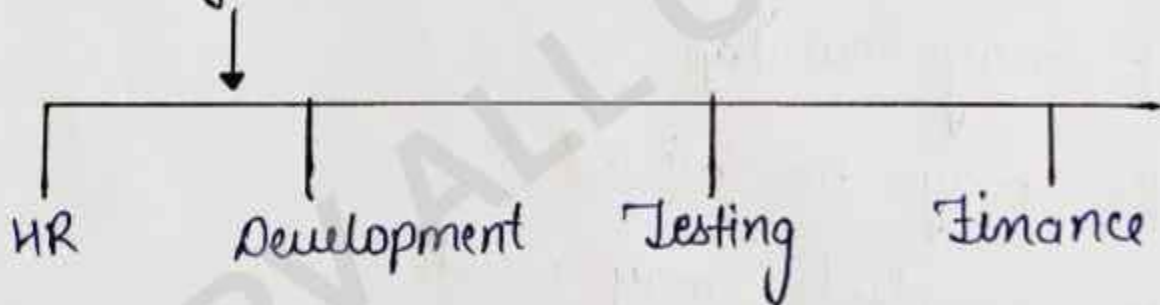
Employees are grouped according to functions or departments.

Example Departments:

- Development
- Testing
- HR
- Finance
- marketing

Each department has its own manager.

Company Head



Characteristics

- Employees report to functional managers.
- Projects are handled within departments.
- Authority remains with department heads.

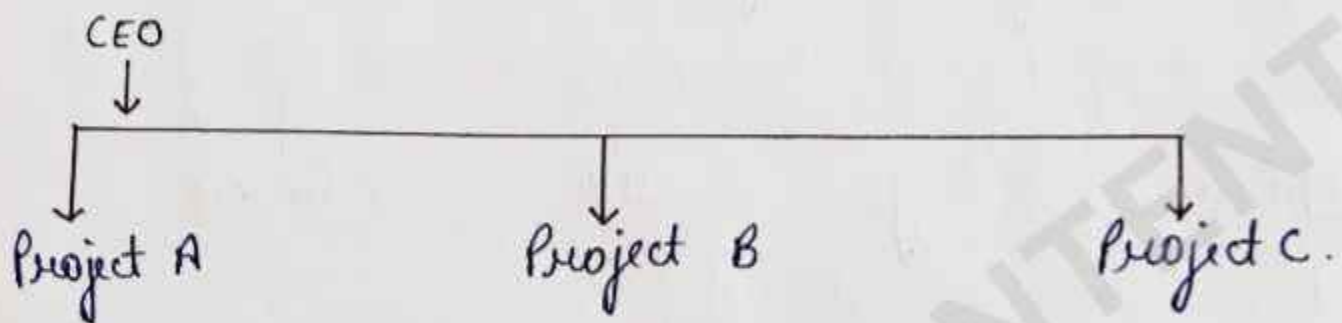
2. Projectized Organization

In this structure

The entire organization is arranged around projects.

A project manager has full authority.

Team members work directly for the project



Characteristics -

- Teams are dedicated to projects
- Project manager controls resources.
- Strong focus on project objectives.

3. Matrix Organization -

Matrix organization is a combination of:

- Functional Organization

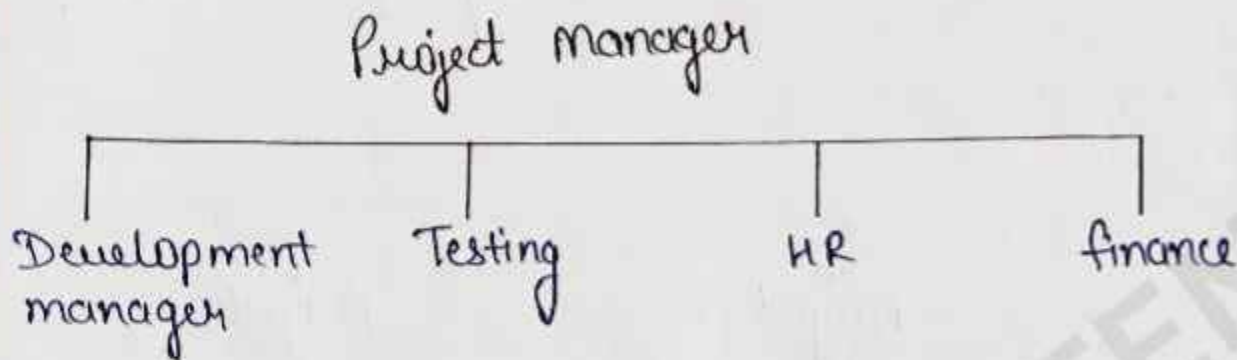
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- Projectized Organization

Employees report to:

- Functional manager and
- Project manager

This is called dual reporting relation-ship.



Characteristics-

- Shared Resources
- Dual Authority
- Flexible structure.

PROJECT ROLES & RESPONSIBILITIES

1. Project manager-

The project manager is the most important person in project management.

Responsibilities-

- Planning
- Team management

- Risk management
- Communication
- Budget management.

2. Project Team members -

These are the people who perform project tasks.

Ex - Developers

Designers

Testers

Engineers

Responsibilities -

- Complete assigned work
- Follow deadlines
- Report Progress
- Collaborate with team members.

3. Functional manager -

It manages a specific department.

Responsibilities

- Provide Technical Guidance
- manages Departmental staff
- Allocates resources
- maintain technical quality.

PROCESS AUTOMATION

Process Automation means using software tools & technology to perform repetitive project tasks automatically with less human effort.

It helps organizations save time, reduce errors, improve speed & increase efficiency in managing projects.

Objectives of Process Automation-

1. Reduce manual work
2. Increase project efficiency
3. Improve accuracy
4. Save time & cost.
5. Speed up project completion

Features of Process Automation

1. Automatic task Execution → Routine tasks are completed automatically by software.

Ex:- Sending reminder emails.

2. Workflow management → Automation controls the flow of work from one stage to another.

Ex:- After approval, the next task starts automatically.

3. Real-time monitoring → managers can track project progress instantly.

Ex:- Dashboard showing completed & pending tasks.

4. Automatic Reporting → Reports are generated without manual calculations.

Ex:- Weekly progress reports.

5. Notification system → Team members receive alerts automatically.

Ex:- Deadline reminders
meeting notification

Components -

1. Input → Data entered into the system.
2. Processing → Software processes the data automatically.
3. Output → Final information generated by the system.
4. Feedback → System checks results & updates information.

Process Automation cycle -

1. Identify Repetitive Tasks
2. Design automated workflow
3. Select automation tools.
4. Implement automation
5. monitor performance
6. Improve the process continuously.

PROJECT CONTROL & PROCESS INSTRUMENTATION

Project control & process instrumentation are important concepts in project management.

They help managers monitor project progress, compare actual performance with planned targets & take corrective actions when needed.

Project Controls → managing & controlling project work to achieve goals.

Process Instrumentation → using tools, techniques & measurements to observe & improve project processes.

PROJECT CONTROL [May 2024]

Project control is the measures process of measuring project performance, comparing it with the project plan identifying deviations & taking corrective actions to keep the project on track.

Objectives of Project Control-

1. Complete project on time
2. Keep project within budget.
3. Maintain required quality.
4. Use resources efficiently.
5. Detect problem early.
6. Improve decision making.

Activities -

1. Planning Baseline → Set original targets for cost, schedule & scope.
2. Performance measurement → Measure actual progress regularly.

3. Comparison → Compare actual performance with planned performance.
4. Variance Analysis → Identifying differences between planned & actual results.
5. Corrective Action → Take steps to solve delays or problems.

Areas of Project Control-

- a) Schedule Control → Ensures project tasks are completed on time.
- b) Cost Control → Controls expenses & budget.
- c) Quality Control → Ensures output meets standards.
- d) Resource Control → manages manpower, tools and materials.
- e) Risk Control → monitors & reduces project risk.

Benefits -

- Better monitoring
- Reduced delays
- Lower & overruns
- Improved productivity
- Better quality results

PROCESS INSTRUMENTATION [May 2024]

Process Instrumentation means using metrics, tools, indicators & automated systems to measure & monitor project processes.

It provides data for better control and improvement.

Objectives of process instrumentation -

1. Measure process performance
2. Detect issues quickly
3. Improve efficiency
4. Support project control
5. Maintain consistency
6. Help in decision making

Instruments / Tools used in Process -

1. Performance metrics → measures productivity, cost, defects, progress, etc.
2. Dashboards → Visual display of project status.

3. Charts & graphs → Gantt charts, bar charts, progress chart.

4. Tracking Software → Project management software for monitoring

5. Reports → Daily, weekly, monthly reports.

6. Alerts & Notifications → Automatic reminders for deadlines or delays.

Common Metrics Used-

- Schedule variance
- Cost variance
- Task Completion rate
- Resource utilization
- Defect rate
- Productivity level

MANAGEMENT INDICATORS [Dec 2024][May 2023]

Management Indicators are measurable values or signals used by managers to evaluate the performance, progress, efficiency and success of an organisation or project.

These indicators help management in planning, controlling, decision-making & improving overall performance.

Management Indicators show whether work is going in the right direction or not.

Definition -

Management indicators are quantitative or qualitative measures used to monitor organizational activities & compares actual performance with planned objectives.

Objectives -

1. measures performance accurately.
2. Track progress toward goals.
3. Identify problems early

4. Support decision-making
5. Improve productivity & efficiency.
6. Ensure proper use of resources.
7. Help in future planning

Importance of Management Indicators-

1. Provide clear performance data.
2. Help managers control operations
3. Detect delays or cost overruns
4. Improve accountability.
5. Increase organizational effectiveness.

Types of Management Indicators-

- 1) Financial Indicators → Used to measure financial performance.

Examples:-

- Profit
- Revenue
- Return on Investment
- Cost variance
- Budget utilization

2. Productivity Indicators → measure efficiency of work & output.

Examples:-

- Output per employee
- Tasks completed per day
- machine utilization rate.

3. Quality Indicators → measure quality of products or services.

Examples:-

- Defect rate
- Customer complaints
- Rework percentages.

4. Time Indicators → measure schedule and deadlines.

Examples:-

- Project completion time
- Delay rate
- Average response time

5. Customer Indicators → measure customer-related performance.

Examples:-

- Customer satisfaction score.
- Retention use

[MIMP]

[May 2022]

[June 2025]

[May, 2023]

[Dec 2024]

[May 2024]

LIFE - CYCLE EXPECTATIONS

Life-cycle expectations refer to the expected outcomes, goals, activities & results at each phase of a project life cycle.

Every project passes through different stages & management expects specific deliverables and progress from each stage.

Basically, life cycle expectations define what should happen during each phase of the project.

Importance of Life Cycle Expectations -

- Provides clear direction for each phase.
- Helps measure progress of the project.
- Reduces confusion among team members.
- Improves planning & control.
- Ensures timely completion of task.
- Helps achieve project goals successfully.

Project Life Cycle Phases & Expectations.

A project usually has four main phases:

1. Inception / Initiation
2. Planning / Elaboration
3. Execution / Construction
4. Closure / Transition

1. Inception / Initiation Phase -

This is the starting phase where the project idea is identified & approved.

Expectations in this Phase

- Define project objectives clearly.
- Identify project scope.
- Analyze feasibility
- Estimate budget & time.

Deliverables

- Project Proposal
- Feasibility report
- Initial schedule
- Budget estimate.

2. Planning / Elaboration Phase

Detailed planning is done before actual work starts.

Expectations in this Phase-

1. Create Project Plan.
2. Divide work into task
3. Allocate resources.
4. Prepare risk management plan.
5. Set deadlines & milestones.

Deliverables-

- Detailed project plan
- Work Breakdown Structure
- Schedule chart
- Resource Plan.

3. Execution / Construction Phase-

Actual project work is performed in this phase.

Expectations in this Phase

1. Complete task as planned
2. Maintain quality of work
3. Use resources efficiently.

4. monitor progress regularly
5. Solve issues quickly.
6. Keep project within budget & schedule.

Deliverables -

- Product or service output
- Progress output / reports
- Performance updates.

4. Closure / Transition Phase

This is the final phase where the project is completed & delivered.

Expectations in this Phase.

1. Final testing & verification
2. Deliver final product to client.
3. Obtain approval from customer
4. Close contacts & accounts.
5. Release project resources

Deliverables -

- Final Product handover
- Completion report
- Lessons learned documents.

PROCESS DISCRIMINANTS ^[may 2023] _[may 2022]

Process Discriminants are the factors or characteristics that help in selecting, modifying & managing the most suitable project process or methodology.

Every project is different, so no one common process can't be applied to all projects.

Process Discriminants help managers decide which process model, tools, controls & management style should be used.

In simple words, process discriminants are the points that differentiate one project from another & guide the management approach.

Definition:-

Process Discriminants are project-specific factors that influence the choice, structure and execution of the project management process.

Need for Process Discriminants-

1. Every project has different requirements.

2. Helps choose the right project life cycle model.
3. Improves planning & control.
4. Reduces risk & failures.
5. Ensures efficient use of resources.

Examples:-

Suppose a company is building hospital software

- High risk
- High quality required.
- Large team
- Sensitive data.
- Fixed deadlines.

So management choose a formal & controlled process with strong testing & documentation.

Advantages

1. Better Planning
2. Proper process selection
3. Lower risk
4. Better team coordination
5. Higher customer satisfaction

2. Helps choose the right project life cycle model.
3. Improves planning & control.
4. Reduces risk & failures.
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Limitations

1. Wrong analysis may select wrong process.
2. Too many factors can create confusion.
3. Conditions may change during project execution.

